

Robust Intrusion Detection System in CAN Bus through Multi-Scale Feature Fusion

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Abstract—Over the past few decades, as vehicles have become increasingly intelligent, the applications of in-vehicle electronic systems have expanded significantly. However, with the growing complexity of vehicle networks, there is an ever-increasing concern for their network security. In particular, the Controller Area Network (CAN) bus has become a critical medium for communication between various Electronic Control Units (ECUs) within a vehicle. Since the design of the CAN bus lacks sufficient security measures, it is vulnerable to various network intrusions.

To address this security challenge, researchers have been searching for ways to enhance the network security of the CAN bus to ensure that vehicle systems are not compromised by unauthorized access or network attacks. This paper introduces a robust intrusion detection system (IDS) for the CAN bus in vehicles, employing a novel Multi-Scale Feature Fusion technique. Leveraging the distinct capabilities of Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and Transformer neural network architectures, the proposed methodology adeptly captures and prioritizes both shallow and deep features of CAN bus data. Besides, it enhances detection accuracy and robustness against various cyber-attacks. Evaluation on the Car-hacking dataset demonstrates superior performance, achieving a precision, recall, and F1-score of 100%. Verified by an ablation study, this approach promises a substantial advancement in safeguarding in-vehicle networks.

Index Terms—Controller Area Network (CAN), Intrusion Detection System (IDS), In-Vehicle Network, Automotive Security

I. INTRODUCTION

In the era of advanced digital transformation, modern vehicles are evolving beyond mere transportation methods, becoming intricate systems of connected networks. This progress is fueled by the expansion of the Internet of Vehicles (IoV) and Internet of Things (IoT) technologies [1]. Within these highly networked vehicles, Controller Area Network (CAN) serves as the central nervous system. It communicates with Electronic Control Units (ECUs) to manage functionalities ranging from engine control to advanced driver-assistance systems.

However, this increasing connectivity brings forth a myriad of security concerns. Although CAN protocol is the backbone of in-vehicle networks (IVNs), it is well known for lack of intrinsic security such as encryption or authentication [2]. This vulnerability makes it possible to exploit the system by injecting malicious messages. It can be seen in different forms such as Denial of Service (DoS), fuzzy, and spoofing attacks. Given these security loopholes, there is a need for Intrusion Detection Systems (IDSs) designed for CAN bus. The system structure of in-vehicle network is shown in Fig.1: it mainly

includes ECU components, external communication facilities and CAN bus.

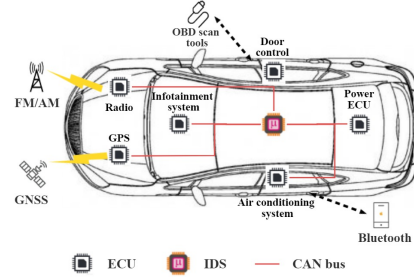


Fig. 1: Basic structure of in-vehicle network.

Machine Learning (ML) and Deep Learning (DL) have gained traction in solving intelligent connected vehicle problems [3]. However, applying them to CAN bus IDS poses unique challenges. One primary issue is the multi-scale nature of CAN bus data, which consists of various types of messages. Each of these messages carries different kinds of information, such as sensor readings, control commands, and system statuses. Traditional ML and DL models tend to concentrate on a single scale or dimension of the data, potentially overlooking complexities captured by multi-scale analysis. Moreover, current IDSs mainly focus on shallow features (e.g., packet frequencies and data payload statistics) or deep features (e.g., sequential patterns), neglecting the potential benefits of fusing these features [4]. It often results in models good at detecting certain types of attacks but performing poorly against novel attack patterns.

To address these issues, this paper introduces a novel approach to intrusion detection in CAN bus utilizing Multi-Scale Feature Fusion. It uses Long Short-Term Memory (LSTM) for sequential pattern recognition, Convolutional Neural Networks (CNN) for local feature identification, and Transformer for solving long-range dependencies problems. By combining these architectures, our model captures both shallow and deep features of CAN bus data. This hybrid approach not only improves detection accuracy of our model, but also makes it robust against a variety of cyber-attacks, as validated on the Car-hacking dataset. The method then integrates these diverse features using a Multi-Scale Feature Fusion technique, bolstered by an attention mechanism to automatically prioritize the most relevant features.

The main contributions of this paper are as follows:

- 1) To extract message features from both spatial and temporal perspectives, this paper introduces a Multi-Scale Feature Fusion approach that integrates output from multiple neural network architectures. Combining CNN, LSTM and Transformer, it offers a more comprehensive feature set for intrusion detection in CAN bus. In contrast to [5], our method takes into account the spatial features of images and also considers the dependency relationships among CAN messages as temporal data.
- 2) Our method utilizes an attention mechanism to automatically prioritize feature sets during the fusion process. It thereby enhances the model's ability to detect various types of cyber-attacks effectively.
- 3) We did a rigorous evaluation of the proposed method using the Car-hacking dataset. Then we compare it against existing state-of-the-art intrusion detection techniques, demonstrating its superiority.

II. RELATED WORK

Various IDSs have been proposed to detect malicious traffic. Among them, there are two main methods: rule-based and ML-based methods. Rule-based methods extract characteristics of normal CAN messages (such as periodicity and frequency of messages with a specific ID, or entropy of CAN bus) and judge messages that do not follow these rules as potentially malicious [6]–[8]. However, this method cannot cover all normal behaviors. For example, in the method of detecting periodicity of messages, some messages are judged as abnormal because the frequency deviates from normal due to blocking delays.

ML and DL techniques have been increasingly applied to intrusion detection tasks in in-vehicle networks. Their capacity to learn directly from data makes them excellent for identifying subtle alterations in vehicle networks. Yang et al. [9] introduced an Attention-LSTM neural network that improved detection accuracy while reducing false alarms. Desta et al. [10] proposed an IDS for in-vehicle networks using CNN trained on recurrence plots generated from CAN frame arbitration IDs, which is called Rec-CNN. Song et al. [11] proposed an IDS based on a Deep Convolutional Neural Network (DCNN) optimized for CAN bus. The DCNN model learns to identify malicious traffic patterns without the need for hand-designed features, offering an improvement over traditional machine-learning algorithms.

These specialized models have advantage at detecting specific attacks but struggle with complex patterns. For example, temporal features (intervals between timestamps) are crucial for identifying DoS attacks. However, when an attacker implements spoofing RPM (Revolutions Per Minute) or spoofing gear attack, the time feature may not be useful at all. This complexity is not only due to different types of attacks occurring simultaneously, but also due to the multi-dimensional nature of the CAN bus data itself. The inherent complexities of attack scenarios on CAN bus necessitate a more sophisticated model. Traditional approaches are often monolithic, focusing on one type of feature—either temporal, localized, or structural. While it is effective in controlled settings, these models cannot

capture interactive features across multiple data dimensions, limiting their attack recognition performance.

III. PROPOSED IDS FRAMEWORK

To address the multifaceted challenges of intrusion detection in CAN bus systems, our methodology utilizes the advantage of LSTM, CNN, and Transformer architectures. Furthermore, we introduce a Multi-Scale Feature Fusion technique, enhanced with attention mechanism to automatically prioritize crucial features during the training phase. The flowchart can be referenced in Fig. 2.

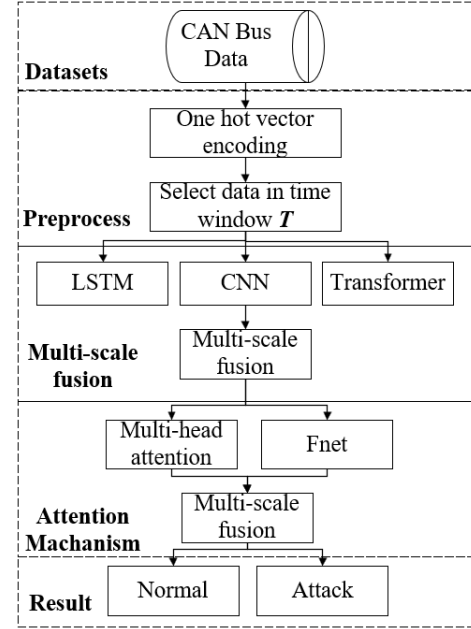


Fig. 2: The flowchart of our proposed IDS.

After undergoing data preprocessing, the collected dataset is formally input into the neural network. Initially, the data passes through three basic network architectures, resulting in feature maps that are concatenated based on their feature dimensions. Subsequently, high-dimensional feature fusion takes place, followed by the selection of weights through the attention mechanism. Finally, classification is performed using fully connected layers. The methodology unfolds as follows:

A. Preprocess

CAN is a protocol used to communicate with ECUs inside the vehicle. Fig.3 depicts the overall architecture of a CAN frame. It is made up of different fields like ID, DATA, Remote Transmission Request (RTR) control, etc. ID and DATA are the most important fields among them as they carry the identity of message sender and effective payloads.

We utilize the CAN ID field. Since CAN IDs are in hexadecimal format, we aim to encode them in binary for better matrix representation. Consequently, we encode each element of the CAN ID in its binary form as a one-hot vector. The one-hot vector encoding ensures that one position is set to 1, while the rest are set to 0. For example, if an



Fig. 3: A standard CAN frame.

element of the CAN ID is “0x2a0,” and “2” is selected, the corresponding one-hot vector will have only the second position as 1, with all other positions as 0. We represent a 3-bit CAN ID with three such vectors, resulting in a 48-bit vector format for “0x2a0” ($16 * 3 = 48$ bits). For instance, if the CAN ID is “0x2a0”, 3 one-hot vectors are concatenated together, like $[0100 \dots 000, 0 \dots 0100000, 1 \dots 0000000]$. Besides, DLC (Data Length Code) field is added, thus each feature gets 49 bits.

Next, we collect messages within a time window. We set the time window as T . In all T messages, if one of them is an attack message, then the label for messages generated within this window is classified as an attack. Otherwise, it is classified as normal. This matrix, generated in this manner, serves as the input X for the neural network.

B. Feature Extraction

1) *LSTM for Sequential Pattern Recognition:* LSTM is used to parse temporal dependencies in CAN bus data. It captures sequential patterns and time-domain dynamics. Given an input sequence $X = [x_1, x_2, \dots, x_T]$ of length of time series T , LSTM updates its states through a series of gating mechanisms. Specifically, at time step t , the LSTM unit updates its hidden state h_t based on previous hidden state h_{t-1} and previous cell state c_{t-1} .

The computations inside the LSTM unit can be described as follows:

$$\begin{aligned}
 f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \\
 i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \\
 \tilde{c}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_c), \\
 c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \\
 o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \\
 h_t &= o_t \odot \tanh(c_t)
 \end{aligned}$$

where f_t, i_t, o_t are activation vectors for the forget gate, input gate, and output gate, respectively, regulating information flow in the LSTM cell. σ is the sigmoid function, normalizing activations to the range $[0, 1]$. W_f, W_i, W_C, W_o are weight matrices corresponding to the forget gate, input gate, candidate cell state, and output gate. b_f, b_i, b_C, b_o are bias vectors associated with the aforementioned gates and candidate state. $\tilde{c}_t$ is the candidate cell state, suggesting potential new information for the current cell state. c_t is the Cell state at time t , storing temporal information. h_t is the hidden state at time t , the LSTM's output for the current timestep. $\odot$ is the Hadamard product, signifying element-wise multiplication.

The LSTM unit is then utilized for each time step t across the entire input sequence. In the context of CAN bus data, each x_t represents a message at time t , and the LSTM could learn to capture important temporal patterns within the message sequence, which are crucial for detecting intrusions.

2) *CNN for Local Feature Identification:* CNNs help extract local features from CAN bus data by identifying spatial patterns within messages. The convolution operation is crucial for finding hierarchical patterns in the data. Since CAN bus data is one-dimensional and can be considered as time-series data, 1D convolution might be relevant.

Considering input data X of shape (T, D) , where T is the length of the time series, D is the number of input channels, and a convolution kernel K of shape (F, D) where F is the filter size, the convolution operation at time t is articulated as:

$$Z_t = f_{CNN}(\sum_{i=0}^{F-1} X_{t+i} \odot K_i) + b$$

where Z_t is the feature value at time t in the resulting feature map Z , and is of shape $(T - F + 1, D')$, where D' is the number of filters (output channels). f_{CNN} is an activation function, such as the Rectified Linear Unit (ReLU), defined as $f_{CNN} = \max(0, x)$. The bias term b is typically a scalar or a vector of shape $(D',)$ that is added to each element in the feature map. $\odot$ is element-wise multiplication.

For intrusion detection in CAN bus systems, CNNs can recognize spatial patterns linked to different cyber-attacks. They generate spatial features, which can be further analyzed. This includes spotting unusual message sequences or abnormal communication patterns between ECUs, which could indicate malicious activity.

3) *Transformer for Long-Range Dependencies:* Transformers, through self-attention mechanisms, manage long-range dependencies effectively. Given an input sequence $X = [x_1, x_2, \dots, x_T]$ of length of time series T , each x_t being a D -dimensional input vector, the Transformer creates output sequence $Y = [y_1, y_2, \dots, y_T]$ via its self-attention mechanism:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

where Q, K, V are weight matrices used for linear transformations to create Query, Key, and Value matrices, respectively. K^T means transposition of K , d_k represents the dimensionality of the query and key vectors, and softmax is applied to the rows of the matrix $QK^T/\sqrt{d_k}$.

The scaled dot-product attention can be extended to multi-head attention, thus it allows the model to focus on different parts of the input sequence for different tasks or reasons:

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W_o$$

where $head_i = Attention(QW_{Qi}, KW_{Ki}, VW_{Vi})$.

Each position (or token) in the input sequence has a corresponding output which is passed through the same feed-forward network. This consists of two linear transformations with a ReLU activation in between:

$$FFN(x) = \max(0, xW_{FF1} + b_{FF1})W_{FF2} + b_{FF2}$$

where W_{FF_1} , b_{FF_1} , W_{FF_2} and b_{FF_2} are the parameters of the feed-forward networks.

The final representation $Y^{Transformer}$ can thus be obtained through a series of transformations and attention mechanisms. It is able to capture both local and global dependencies within the input sequence X .

In the context of CAN bus data, the Transformer can be extremely valuable in identifying complex attack patterns by considering the full context of input messages. Integrating such representations with proposed Multi-Scale Feature Fusion methodology, it could recognize sophisticated cyber-attacks on in-vehicle networks capturing both local and global contextual information from in-vehicle network data.

C. Multi-Scale Fusion

In DL, concatenation is a commonly used operation that is typically employed to concatenate multiple tensors along a particular dimension, creating a larger tensor. The advantage is that it allows for the aggregation of the highest-order features to construct a comprehensive new representation. This operation is usually performed along the last dimension of the tensors, often the feature dimension. It implies that tensors must have the same size in all dimensions except the one being concatenated.

This operation finds widespread application in neural networks across various scenarios, with one of the most common being the fusion and connection of information between layers or different branches. In this model, we extract the highest-order features generated from top dense layers of the base model. We then employ concatenation to integrate all these features into a new concatenated layer containing all the extracted features.

The extracted feature representations from LSTM, CNN, and Transformer, and are denoted as H^{LSTM} , Z^{CNN} , and $Y^{Transformer}$ respectively. They are then concatenated to produce a unified feature vector, given by:

$$F = Concat(H^{LSTM}, Z^{CNN}, Y^{Transformer})$$

D. Attention Mechanism

To dynamically allocate weights to various feature segments in F and emphasize segments pertinent to intrusion detection, the attention weight α_i for the i -th feature is computed via:

$$\alpha_i = \frac{\exp(e_i)}{\sum_{k=1}^N \exp(e_k)}$$

$$e_i = V_a^T \tanh(W_a F_i + b_a)$$

where W_a is weight matrix, b_a is bias vector, F_i is the i -th feature in F , and N is the total number of features in F . In our experiment, we employ two main types of attention mechanisms. One is the multi-head attention mechanism, which is explained previously. The other is the Fourier Neural Network (FNet) [12]. FNet is based on Discrete Fourier Transform (DFT) and it leverages DFT with the following formula:

$$X(k) = \sum_{n=0}^{N-1} x_n e^{(-\frac{2\pi i}{N}nk)}, 0 \leq k \leq N-1$$

where x_n is the original sequence with length of N . For each k , the DFT generates a new representation X_k as a sum of all of the original input tokens x_n .

Essentially, FNet replaces the self-attention sub-layer of each Transformer encoder layer with a Fourier sub-layer, and the formula for this transformation is as follows:

$$F = \Re(\mathcal{F}_{seq}(\mathcal{F}_h(x)))$$

where $\Re$ means keeping the real part of the result, $\mathcal{F}_h(x)$ means applying 1D DFT along the hidden dimension and $\mathcal{F}_{seq}$ along the sequence dimension.

Next, the fused features F go through these two attention mechanisms subsequently obtain new features $F^{multi-head}$ and F^{FNet} . Consequently they are concatenated using the concatenation method described previously. Feature vector F' is computed as:

$$F' = Concat(F^{multi-head}, F^{FNet})$$

E. Classification Layer

Finally, F' is passed to a classification layer, utilizing a softmax function to calculate the probability of each class:

$$P(y|F') = \frac{\exp(W_y F' + b_y)}{\sum_{c=1}^C \exp(W_c F' + b_c)}$$

The approach provides a robust and comprehensive methodology for effectively detecting intrusions in CAN bus systems. By integrating various neural network architectures and judiciously applying a fusion technique, the model assures enhanced detection accuracy and robustness against a spectrum of cyber-attacks.

IV. EXPERIMENT AND PERFORMANCE

A. Data Preparation and Model Setup

a) *Dataset*: The dataset employed in this study is centered on in-vehicle network security, specifically focusing on the CAN bus. It forms the communication backbone within an automobile. It is known as the ‘‘Car-hacking-dataset’’, which was released by the Hacking and Countermeasure Research Lab and can be obtained from reference [14]. The dataset is imbued with instances that are emblematic of normal network traffic and various cyber-attacks, specifically, Denial of Service (DoS) attacks. Every instance in the dataset symbolizes a message transmitted over the CAN bus, encapsulating information such as the Arbitration ID and DLC. In light of maintaining the intrinsic characteristics whilst ensuring computational efficiency, the dataset was pruned to 100,000 instances. A strategic partition was then employed to create: training set with 7030 sequences and testing set with 782 sequences.

b) *Model Architecture*: In order to address the multi-faceted and multi-scale nature of CAN bus data, the proposed model leverages the merits of different neural network architectures, specifically LSTM, CNN, and Transformer blocks, each good at handling different aspects of the data.

- **LSTM Block**, with their advantage in modeling sequential dependencies, form a crucial component of the architecture. The LSTM block within the model consists of

a Bidirectional LSTM layer with 64 units, followed by a dense layer consisting of 100 neurons. This block processes sequences of CAN bus messages, effectively capturing temporal patterns and dependencies within the data.

- **CNN Block**, renowned for their spatial feature extraction capabilities, are employed to scrutinize local patterns within the CAN bus messages. The CNN block comprises three convolutional layers, each followed by max-pooling, and two dense layers. Specifically, the convolutional layers utilize 32, 64, and 64 filters respectively, with a kernel size of 3 and “relu” activation. The succeeding dense layer employs 64 neurons, and the final dense layer utilizes 100 neurons.
- **Transformer Block** captured both local and global dependencies within the sequences. This block is characterized by an embedding dimension of 49, utilizing 2 attention heads and a feedforward network hidden layer size of 32 within the Transformer. This block, through self-attention mechanisms, ensures that the model is cognizant of both local and global contextual information within the sequences.

c) Evaluation: The proposed model was evaluated through a variety of metrics and visualizations, ensuring a comprehensive understanding of its predictive capabilities.

- **ROC-AUC:** The Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) was employed, with the model achieving an outstanding AUC score of approximately 0.9999, indicating exceptional discriminatory power between “Normal” and “Attack” classes. The AUC curve is depicted in Fig. 4.
- **Classification Report:** Precision, recall, and F1-score were scrutinized, with the model demonstrating excellent predictive capabilities, achieving a score of 1.000 across these metrics for both classes. Moreover, an overall accuracy of 1.000 further reinforced the model’s advantage in intrusion detection.
- **Confusion Matrix:** The confusion matrix showcased the model’s superior classification capabilities, with no instances of mis-classification observed, reflecting its reliability and precision in handling intrusion detection in the in-vehicle network. The confusion matrix for classification on all samples is presented in Fig. 5.

B. Model Training and Results

In this section, we delve into the experimental results obtained from various models employed in this study, inclusive of the proposed model, which amalgamates the LSTM, CNN, and Transformer architectures under the umbrella of a Multi-Scale Feature Fusion technique.

In order to enhance the credibility of our proposed IDS, we also compared it with some classical ML methods, such as Decision Trees (DT), Random Forest (RF), XGBoost, and Logistic Regression (LR). Furthermore, the research presented in [13] also reports the classification outcomes achieved using GIDS on this dataset. The result is presented in Table I.

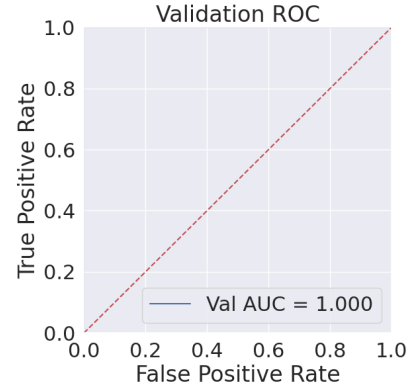


Fig. 4: The AUC curve of our proposed model.

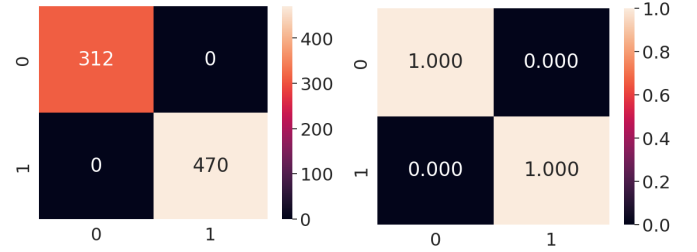


Fig. 5: The confusion matrix heatmap of the proposed method.

TABLE I: Comparison with other ML methods.

Method	Accuracy(%)	Precision(%)	Recall(%)	F1
DT	88.11	86.14	83.65	0.8488
RF	94.12	98.54	86.54	0.9215
XGBoost	99.10	100.00	97.76	0.9887
LR	99.36	100.00	98.40	0.9919
LSTM	99.87	99.68	100.00	0.9984
CNN	98.34	100.00	95.83	0.9787
Transformer	99.87	99.68	100.00	0.9984
GIDS (Average)	97.53	97.63	98.65	0.9814
Our method	100.00	100.00	100.00	1.0000

C. Ablation Study

In an effort to dissect the contribution of each neural network architecture (LSTM, CNN, and Transformer) and the attention mechanism within our proposed model, we conducted an ablation study. This involved systematically ablating (removing) one component at a time and observing the impact on model performance.

a) Removal of FNet attention mechanism: FNet is designed as an auxiliary mechanism to enhance the stability of multi-head attention mechanism. To explore its impact on detection results, we first remove this part to observe the changes. The confusion matrix for the model without FNet (where attention mechanism consists solely of multi-head attention) in the detection results is shown in Fig. 6. It indicates that removing FNet does not significantly impact results as shown in figures.

b) Removal of both attention mechanism: To further investigate the influence of the entire attention mechanism,

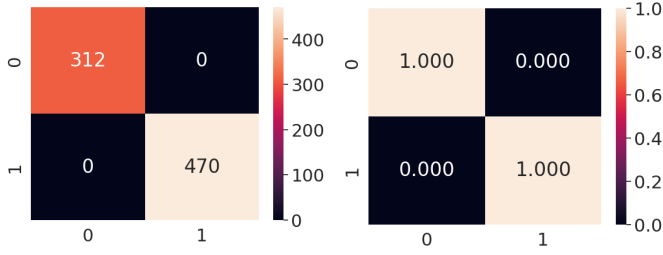


Fig. 6: The confusion matrix heatmap of the proposed method without FNet (only consisting of multi-head attention).

we subsequently remove the multi-head attention mechanism. The confusion matrix results are shown in Fig. 7.

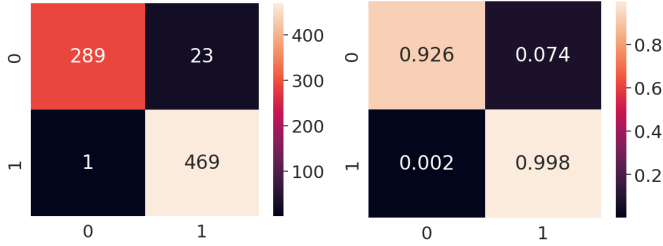


Fig. 7: The confusion matrix heatmap of the proposed method without FNet and multi-head attention.

The result of this ablation study is shown in Table II. From the experiments conducted above, we can draw following conclusions:

- 1) In the absence of an attention mechanism, the model's performance with only feature fusion is comparable to the results of single network architectures. For instance, using only the Transformer achieves an accuracy of 99.87%, and even the worst-performing single model, CNN, achieves an accuracy of 98.34%. This is only 1.66% lower than the feature fusion model based on attention mechanisms.
- 2) When an attention mechanism is used, introducing an additional type of attention mechanism does not significantly improve the accuracy of our model. This might be due to the relatively simplicity of the dataset. The effectiveness of attention mechanisms needs further exploration in subsequent experiments.

TABLE II: Comparison in ablation experiments.

Multi-head	FNet	Accuracy(%)	Precision(%)	Recall(%)	F1
✓	✓	100.00	100.00	100.00	1.0000
✓	✗	100.00	100.00	100.00	1.0000
✗	✗	96.93	92.63	99.66	0.9602

V. CONCLUSION

This work introduced a novel intrusion detection method for in-vehicle CAN bus systems, leveraging the capabilities

of LSTM, CNN, and Transformer networks through a Multi-Scale Feature Fusion technique. The proposed model exhibited superior performance, outshining several traditional and deep learning models, and proving its efficacy in identifying a variety of cyber-attacks with high precision, recall, and F1-score of 100%.

The ablation study further revealed the significance of each neural network architecture and the attention mechanism, substantiating their individual contributions. As in-vehicle technologies continue to evolve, future work will delve into further optimizing the fusion technique, exploring real-world applicability thereby solidifying its stance as a robust IDS in the realm of in-vehicle network security.

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